Weakly Supervised Pneumonia Detection and Localization on Chest X-rays using a CNN and Class Activation Maps

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1. Problem Statement

Chest X-rays are the standard tool for diagnosing pneumonia, yet hospital records usually store only the final diagnosis (pneumonia or not), not the pixel location of the lung opacity. Standard supervised localisation (e.g. R-CNN) needs bounding-box labels that most datasets lack. Our goal is to (i) classify a chest X-ray as *pneumonia* or *normal* and (ii) **localise** the pneumonia region with a bounding box while training only on binary image-level labels, i.e. **weakly supervised localisation**. We re-implement and evaluate the approach of Huang, Cortes and Monam (*Weakly Supervised Pneumonia Localization*).

2. Dataset

We use the Kaggle **RSNA Pneumonia Detection Challenge** (stage 2) data: DICOM chest X-rays (1024×1024) with a class label (*Lung Opacity*, *Normal*, *No Lung Opacity / Not Normal*) and ground-truth boxes for pneumonia cases. As in the paper, the ambiguous *Not Normal* class is removed to keep the task clean. The resulting set has **14,863 images: 6,012 pneumonia (40.4%) and 8,851 normal (59.6%)**. It was split **70/20/10** (stratified) into train, validation and test (1,487 test images). Images are resized to 128×128 and normalised per pixel (training-set mean and standard deviation). Ground-truth boxes are used only for evaluation, never for training.

3. Approach



Fig. 1 – Pipeline: classify with a CNN, then reuse the classifier's own weights to localise.

Classification. A 10-layer CNN (3×3 convolutions with zero padding, ReLU and batch-norm; three max-pools) is followed by a **Global Average Pooling** (GAP) layer and a single fully-connected (FC) layer. This exact structure is required by CAM. The input has two channels: the original image and a lung-segmented copy (image × lung mask), giving the network a hint of where the lungs are. Training used Adam (lr 1e-4), batch size 64, 20 epochs, cross-entropy loss, and the epoch with the best validation accuracy was kept. LR, SVM and Random Forest on flattened 32×32 images serve as baselines.

Localisation (CAM). For an image predicted as pneumonia, the Class Activation Map is the sum of the last convolutional feature maps weighted by the FC weights of the pneumonia class: $M(x,y) = \Sigma$ over k of $w[k] \cdot f[k](x,y)$, where f[k] is the k-th feature map and w[k] its pneumonia-class weight. The map is up-sampled and scaled to [0,1], thresholded (0.45), and non-zero pixels are grouped into connected regions with a depth-first search (DFS). Each region gets a bounding box (min/max x,y), and boxes whose area lies beyond 2 standard deviations are discarded. Localisation quality is measured by **IoU** = $|A \cap B| / |A \cup B|$ between predicted and ground-truth boxes.

4. Implementation Overview

The project is written in Python/PyTorch and trained on a free Kaggle Tesla T4 GPU. Modules: `prepare_data.py` (DICOM → 128×128 cache), `dataset.py` (splits, normalisation, lung mask, 2-channel tensors), `model.py` (CNN+GAP+FC), `train.py`, `baselines.py`, `cam.py` (CAM, DFS clustering, boxes, IoU), `evaluate.py` and `demo.py` (heat-map + boxes for one X-ray). The README gives set-up and run instructions. The paper's U-Net lung segmenter was replaced by a simple Otsu-threshold lung mask, and the supervised R-CNN benchmark was not re-implemented (its published IoU is quoted instead).

5. Results

Classifier	Train acc.	Test acc.
Logistic Regression	87.88%	83.32%
SVM	91.79%	88.70%
Random Forest	100.00%	87.96%
CNN (ours)	**–**	**89.24%**

Confusion matrix (test set; rows = actual, columns = predicted): TN = 811, FP = 75, FN = 85, TP = 516. This gives sensitivity 85.9% (516/601) and specificity 91.5% (811/886).

Localisation (IoU)	Value
Ours: CNN + CAM (threshold 0.45, 516 correctly detected pneumonia images)	**0.2167**
Paper: CNN + CAM (weakly supervised)	0.1508
Paper: supervised R-CNN benchmark	0.1266

Caveats. Our IoU compares the union of predicted boxes with the union of ground-truth boxes and is computed only on correctly detected pneumonia images, so it is not strictly comparable with the paper's figure. Our dataset is also less balanced (40% pneumonia) than the paper's. The 89.2% accuracy is below the paper's 92.5%, and the CNN is only slightly better than SVM/Random Forest (Random Forest also over-fits: 100% train vs 88% test). The CNN's real advantage is that it supports localisation through CAM, which the baselines cannot provide.

6. Conclusions

- A CNN trained with only image-level labels can both detect pneumonia (89.2% test accuracy, 85.9% sensitivity) and localise it (mean IoU 0.217), with no bounding-box supervision.
- A GAP + single FC layer design makes CAM possible at almost no extra cost; the paper reports that large filters in the last layer give imprecise boxes, so we kept all filters at 3×3.
- IoU remains modest in absolute terms: pneumonia boundaries are diffuse, the CAM resolution is coarse, and the lung mask is a heuristic.
- **Future work:** train a U-Net lung segmenter, tune the CAM threshold on the validation set, use higher-resolution inputs and pre-trained backbones, and benchmark against a trained R-CNN.